Open-Source Hardware

The Arduino platform is itself pretty useful for microcontroller projects, but that alone is not enough to propel the popularity and widespread adoption of the platform. Instead of closing the design of the interface board and development environment, the entire Arduino project is deeply entrenched in the emerging practice of open-source hardware. Unlike open-source software, of which Linux is usually the often-cited example, open-source hardware seeks collaboration where physical objects are the outcome. It engages a distributed model of hardware development with contributors generally residing in different parts of the world. Rather than closed systems, open source projects allow an individual freedom to access the source files of a design, make improvements, and redistribute these improvements to a larger community.

The Arduino ecosystem fundamentally embodies this aspiration for openness in design, architecture, collaboration, and philosophy. You can see it for yourself as all of the design files, schematics, and software are freely available to download, use, modify, remake, and even resell. What started as a seemingly serendipitous decision to open the Arduino design and software to the greater community, spurred by the closing of the design school where the Arduino team was first formed, has lead to an entirely new movement in design. The practice of contributors having the liberty to use these designs freely (free as in speech) and with no obligation to buy anything (free as in beer) helps make the Arduino as endearing as a collection of silicon and copper can be. Not to mention that this creative feedback loop ensures that every inspired innovation derived from the Arduino platform is met with ever more imaginative uses for even more new things.

Community

Maybe even more important than the hardware platform itself, the Arduino community is most likely the single greatest cog in the Arduino machine. This community of makers has contributed to the Arduino ecosystem by developing code and libraries, designing new hardware, teaching workshops and classes, and sharing what they've made. The Arduino is now being taught in high schools, colleges, and universities everywhere and "Arduino Night" is a regular ritual at any of a number of hacker spaces around the world. People coming together around the Arduino, as shown in the image of a soldering workshop at Maker Faire in Figure 1-4, contributes to this global community that has made the Arduino such a success.